

Mark Peng

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EDUCATION

University of Toronto

Toronto, Ontario, Canada

Honours B.S. in Computer Science (co-op), Minor in Math & Statistics – GPA: 3.88/4.0

Sep. 2024 – May 2028

- **Relevant Courses:** Software Design, Systems Programming, Data Structures and Algorithms, Machine Learning

TECHNICAL SKILLS

Languages: C, C++, Go, Python, Kotlin, Java, JavaScript, TypeScript, Ruby, SQL, Lua

Frameworks: React, Node.js, Next.js, React Native, Arduino, Django, Express.js, Flask

Developer Tools: Docker, Git, GitHub, GitLab, VS Code, Jira, Postman, Unix, Claude Code, Codex, IntelliJ

Technologies & Concepts: AWS, GraphQL, MongoDB, PostgreSQL, Redis, REST APIs, Agile, Microservices, MCP

EXPERIENCE

Block (Square & Cash App)

May 2026 – Aug. 2026

Software Engineering Intern – Agentic Commerce

Remote, United States

- Led architecture migration retiring a **62K+** item **DynamoDB** table from Block's agentic commerce stack for **Java/Kotlin** services and **gRPC** APIs supporting **10M+** monthly R/W requests, deduplicating upstream logic
- Built scalable merchant discovery for **20K+** sellers using **AWS OpenSearch** and batched cross-service **RPC** APIs
- Architected extensible **Java/Kotlin** and **REST** adapter implementing the **Universal Commerce Protocol (UCP)**, establishing service boundaries between Square payments and external platforms, in partnership with **Google Retail** and **Google AI Mode**
- Implemented brand-new payment and tokenization interface, targeting **\$100M+** **aggregate seller GPV**, creating Square's ledger infrastructure for agentic checkouts
- Engineered lightweight **Go** automations in **Cloudflare Worker** images for autonomous edge agents interfacing with 3rd-party enterprise platforms, saving **≈6 hrs/wk/user**, reducing time spent on repetitive tasks

Pulsenics

May 2025 – Aug. 2025

Full-stack Developer Intern

Toronto, Canada

- Refactored **2K+** LOC of hardware-coupled **C firmware** into logic-based modular functions and interfaces, enabling the creation of additional **Ceedling** unit tests with mocks and stubs, raising repository test coverage by **300%**
- Reduced manual verification time by **≈90%** and cut release cycle time by 30% by building a full-stack **Python QA** pipeline for post-production firmware, automating **Modbus** tests, real-time logging, and data validation
- Minimized communication errors from **10% to 0.001%** by implementing a custom ACK protocol in **C** over Ethernet, and optimizing performance through DMA tuning and clock synchronization

Abundant Science

Feb. 2025 – Apr. 2025

Software Developer Intern

Toronto, Canada

- Built **React Native** mobile app with camera API for PHIPA-compliant lateral flow test capture and analysis
- Engineered **≈85%** classification accuracy proof-of-concept **TensorFlow/OpenCV** model in **Python** for rapid tests
- Achieved **<3MB** bundle size and **<100ms** inference latency for on-device classification model by optimizing **TensorFlow Lite** pipeline and applying model quantization

PROJECTS

🔗 reels-cli | C, Python, Bash

August 2025

- Engineered a complete terminal-based media player with a custom **C** video streaming engine and **Python** backend

🔗 ProportionAI | Next.JS, React, MongoDB, Terraform, Gemini

January 2025

- Developed web-app using **Next.JS**, **React**, and **MongoDB** for an AI-powered time insight study app
- Deployed using **Terraform** through an **AWS EC2** deployment to host the study platform in a declarative manner

AWARDS & LEADERSHIP

Top 8 finalist of 450+ participants @ Hack the 6ix – *fine-tuned Qwen3.5-9B to 2.3x perf. on ExploitBench* July 2026

Top 32 of 1000+ participants @ Hack the North, finalist for Warp – *built an AI voice agent delivery app* Sep. 2025

UTMIST AI Open Source Developer – *team focused on contributing to AI open source projects* May 2025

University of Toronto Scholar Award (worth \$10000), Dean's List Scholar – *for positive academic standing* 2024-2026